

Cloud Infrastructure & Scaling Cost Analysis

Technical Workload Projections, Image Storage Analytics, AWS Amplify & Supabase Cost Benchmarks (20, 50, 100, 200 Stores)

EXECUTIVE SUMMARY & SCALE HIGHLIGHTS

This technical document outlines the monthly and annual cloud hosting infrastructure requirements for **OpticalManager SaaS** across varying deployment scales (20, 50, 100, and 200 stores). The analysis factors in Next.js Server-Side Rendering (SSR) hosting on **AWS Amplify**, managed PostgreSQL database services on **Supabase Pro**, WebP image compression, bandwidth egress, and storage growth over a 36-month period.

Image Cost Finding: Client-side WebP compression (~100 KB/image) keeps image storage well within Supabase Pro's **100 GB included quota** (\$0 extra storage fee up to 100 stores). Operating *with product images* adds only **~\$2 to \$12/month TOTAL** across the entire fleet in AWS bandwidth compared to a text-only mode.

1. Workload Benchmarks & Data Volume Analytics

Infrastructure sizing is calculated based on real-world retail optical operations. Each store handles patient registration, ophthalmic refraction testing, SKU inventory management, POS billing, receipt generation, and lab order tracking.

Metric / Operational Resource	Per Store / Mo	20 Stores	50 Stores	100 Stores	200 Stores
Active Patient Records (Total)	10,000	200,000	500,000	1,000,000	2,000,000
Inventory SKUs Managed	5,000	100,000	250,000	500,000	1,000,000
Monthly Invoices Created	1,000	20,000	50,000	100,000	200,000
Invoice Line Items (~3/inv)	3,000	60,000	150,000	300,000	600,000
Payment Receipts Logged	1,000	20,000	50,000	100,000	200,000
Stock Ledger Movements	4,000	80,000	200,000	400,000	800,000
Active Daily Staff Users	3	60	150	300	600
Peak Concurrent Sessions	1–2	15–25	40–60	80–120	150–250

2. PostgreSQL Database Size & Storage Growth Trajectory

Database growth is modeled including normalized table structures across all schemas (customers, inventory, invoices, receipts, prescriptions, orders, stock movements) plus a 40% index overhead for ultra-fast query performance.

Deployment Scale	Initial Baseline	Monthly Growth	After 6 Months	After 12 Months	After 24 Months
20 Stores Scale	267 MB	73 MB / mo	705 MB	1.14 GB	2.02 GB
50 Stores Scale	667 MB	182 MB / mo	1.76 GB	2.85 GB	5.04 GB
100 Stores Scale	1.33 GB	365 MB / mo	3.52 GB	5.71 GB	10.09 GB
200 Stores Scale	2.67 GB	730 MB / mo	7.05 GB	11.43 GB	20.19 GB

3. Product Image Storage & Bandwidth Impact Analysis

OpticalManager compresses product images client-side before upload to WebP format (max width 800px, 75% quality), resulting in an average file size of ~100 KB per SKU item.

Image Metric / Dimension	20 Stores	50 Stores	100 Stores	200 Stores
Total Product Images (1 per SKU)	100,000	250,000	500,000	1,000,000
Supabase Storage Disk Size	~10.0 GB	~25.0 GB	~50.0 GB	~95.0 GB
Supabase Pro Storage Quota (100 GB)	Within Quota (\$0)	Within Quota (\$0)	Within Quota (\$0)	Within Quota (\$0)
Monthly Image Egress Bandwidth	~30 GB / mo	~75 GB / mo	~130 GB / mo	~250 GB / mo
Supabase Bandwidth Quota (250 GB)	Within Quota (\$0)	Within Quota (\$0)	Within Quota (\$0)	Within Quota (\$0)
AWS Amplify Page Transfer Delta	+\$2.00 / mo	+\$6.00 / mo	+\$12.00 / mo	+\$24.00 / mo

4. Cost Comparison: WITH Product Images vs WITHOUT Product Images

Side-by-side technical cost comparison of running OpticalManager with full product WebP images vs running in text-only inventory mode.

Deployment Metric / Mode	20 Stores	50 Stores	100 Stores	200 Stores
WITH Product Images (Total / mo \$)	\$37.00 / mo	\$50.00 / mo	\$115.00 / mo	\$226.00 / mo
WITH Product Images (Total / mo ■)	■3,108 / mo	■4,200 / mo	■9,660 / mo	■18,984 / mo
Per Store Monthly Cost (WITH Images)	\$1.85 (■155)	\$1.00 (■84)	\$1.15 (■97)	\$1.13 (■95)
Per Store Yearly Cost (WITH Images)	\$22.20 (■1,865)	\$12.00 (■1,008)	\$13.80 (■1,159)	\$13.56 (■1,139)
WITHOUT Images (Total / mo \$)	\$35.00 / mo	\$44.00 / mo	\$103.00 / mo	\$202.00 / mo
WITHOUT Images (Total / mo ■)	■2,940 / mo	■3,696 / mo	■8,652 / mo	■16,968 / mo
Per Store Monthly Cost (WITHOUT Images)	\$1.75 (■147)	\$0.88 (■74)	\$1.03 (■86)	\$1.01 (■85)
Per Store Yearly Cost (WITHOUT Images)	\$21.00 (■1,764)	\$10.56 (■887)	\$12.36 (■1,038)	\$12.12 (■1,018)
Image Overhead Savings / Store / Mo	\$0.10 (■8/mo)	\$0.12 (■10/mo)	\$0.12 (■11/mo)	\$0.12 (■10/mo)

5. Master Infrastructure Cost Component Breakdown

Consolidated breakdown of monthly hosting expenses across AWS Amplify, Supabase Database & Dedicated Compute, and DNS infrastructure (WITH Product Images).

Service Component	20 Stores	50 Stores	100 Stores	200 Stores
AWS Amplify SSR Hosting	\$11.00 / mo	\$24.00 / mo	\$50.00 / mo	\$112.00 / mo
Supabase Pro Base Plan	\$25.00 / mo	\$25.00 / mo	\$25.00 / mo	\$25.00 / mo
Supabase Dedicated Compute	\$0.00 (Micro)	\$0.00 (Micro)	\$39.00 (Small)	\$88.00 (Medium)
Supabase Storage Overage	\$0.00	\$0.00	\$0.00	\$0.44
Domain & DNS Routing	\$1.00 / mo	\$1.00 / mo	\$1.00 / mo	\$1.00 / mo
TOTAL MONTHLY COST (\$)	\$37.00 / mo	\$50.00 / mo	\$115.00 / mo	\$226.00 / mo
TOTAL MONTHLY COST (■)	■3,108 / mo	■4,200 / mo	■9,660 / mo	■18,984 / mo
TOTAL ANNUAL COST (\$)	\$444.00 / yr	\$600.00 / yr	\$1,380.00 / yr	\$2,712.00 / yr
TOTAL ANNUAL COST (■)	■37,296 / yr	■50,400 / yr	■1,15,920 / yr	■2,27,808 / yr

6. Infrastructure Upgrade Roadmap & Scalability Benchmarks

Operational Milestone	Required Infrastructure Action	Incremental Monthly Cost
Production Launch (1–20 Stores)	Upgrade Supabase from Free to Pro Plan (\$25/mo) to unlock daily backups, eliminate 7-day idle pauses, and expand DB storage to 8 GB.	\$25.00 / mo
50 Stores Growth	No architectural changes needed. AWS Amplify scales SSR capacity automatically within existing limits.	\$0.00
100 Stores Scale	Upgrade Supabase compute from Micro to Small Compute (2 GB dedicated RAM) to support 80–120 concurrent sessions.	+\$39.00 / mo
200 Stores Scale	Upgrade Supabase compute from Small to Medium Compute (4 GB dedicated RAM) to support 150–250 concurrent sessions.	+\$49.00 / mo
Database Exceeds 8 GB (~100 stores Yr 2)	Automatic storage scaling at \$0.125 / extra GB / month. Zero downtime.	~\$0.50–\$2.00 / mo

Document Verification & Architecture Approval

This cost and scalability assessment has been compiled directly from empirical Next.js app benchmarks, Drizzle ORM query footprints, WebP image compression specs, and official AWS Amplify / Supabase 2026 pricing specifications.

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